Data Science & AI

2024

Algorithms

Analysis of Algorithms

Lecture No.- 01

By- Aditya sir

Topics to be Covered

- **Topic** Schedule
- **Topic** Outcomes
- **Topic** Prerequisites & Books

ABOUT ME

Hello, I'm Aditya.

1. Represented college as the first Google DSC Ambassador.
2. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
3. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored working professions in field of Data Science and Analytics
11. Have been mentoring GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Topic : Lecture Schedule

DAA
↳ Design & Analysis of Algorithms

1. **Analysis of Algorithms**
   1. Algorithm Concept and Lifecycle
   2. Analysis of Algorithms
   3. Methodology & Types of Analysis
   4. Asymptotic Notations
   5. Framework for Analysing Recursive Algorithms
   6. Apriori analysis of Non-Recursive Algorithms → loop/iterative
   7. Analysing Loops
   8. Space Complexity
   9. Mathematical Background

Topic : Lecture Schedule

2. Divide & Conquer (Design Strategies)

1. General Method
2. Max-Min Problem
3. Merge Sort
4. Binary Search
5. Quick Sort
6. Matrix Multiplication
7. Long Integer Multiplication (LIM)
8. Master Method for D and C Recurrences
9. Recursion Tree

Framework

Shortcut

Analysis

Design strategies

1) Divide & Conquer

2) Greedy

3) Dynamic Programming (DP)

graph

Topic : Lecture Schedule

3. **Sorting Algorithms**
   1. Basic terminology → Classification
   2. Methods
      1. Bubble Sort
      2. Selection Sort
      3. Insertion Sort
      4. Radix Sort

Code
Example
Time Complexity

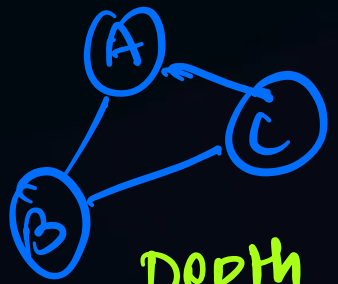
→ Example + Questions

PYQs
Imp Practice Questions.

Topic : Lecture Schedule

4. Graph Algorithms

1. Representation of Graphs → Directed / undirected, weighted / unweighted
2. Graph Traversals



DFS — Depth First Search

1. Undirected Connected Graphs
2. Undirected Disjoint Graphs: DFT
3. Directed Graphs & Types of Edges
4. DAG → Directed Acyclic Graph → Topological sort.

BFS — Breadth First Search

1. FIFO BFS
2. LIFO BFS

* Graph → MCST (Minimum Cost Spanning Tree)

(Greedy)

→ Prims
→ Kruskal
* → Dijkstra

} MCST Construction.

* Graph → Shortest Path Algorithms

→ Single Source Shortest paths (SSSP)

1) Dijkstra's SSSP Algo (greedy)
2) Bellman Ford SSSP Algo (Dynamic Prog)

→ All Pairs Shortest Paths (APSP)

1) Floyd Warshall Algo.

Topic : Lecture Schedule

5. **Heap Algorithms**
   1. Operations : Create, Insert, Delete, Modify
   2. Applications : Heapsort

Kruskal

Topic : Lecture Schedule

6. **Sets** → * Kruskal (Set + Heap)
   1. Representations
   2. Operations
      ↳ {Union, Find}

7) Miscellaneous :-

1) Reference Books :

★ a) Introduction to Algorithms.
→ Thomas Cormen.

b) Fundamentals of Algorithms
→ Horowitz & Sahni

Outcomes :

→ GATE (other relevant Exams)
→ Semester Exams (if in College)
→ Company Placements ✓
→ Interviews ✓

6 months ★

30th Jan → Amazon Coding Test (200+) ✓
↓
3rd Feb → GATE exam ✓
↓
5th Feb → Amazon Interview ✓

4th Feb

* Pre-requisites :-

1) Basic Programming Concepts
   - → if-else
   - → loops
   - → etc

2) Data Structure
   - → (Stack, Queue, Tree, etc)

*3) Basic maths
   - ↳ Series (AP, GP,)
   - ↳ Basic Probability
   - ↳ Log

Algorithms

- Artificial Intelligence (AI)
- Fundamentals of machine Learning (ML)
- Deep Learning (DL)
- NLP (Natural Language Processing)
- Research work (college + Industry)

* Analysis of Algorithms :

→ what is an Algorithm ?

↳ An algorithm consits of a finite set of instructions / steps to solve a particular problem.

→ Sequential

→ unambiguous

↳ fundamental instructions

(atomic)

$a = b + c$ → fundamental operation

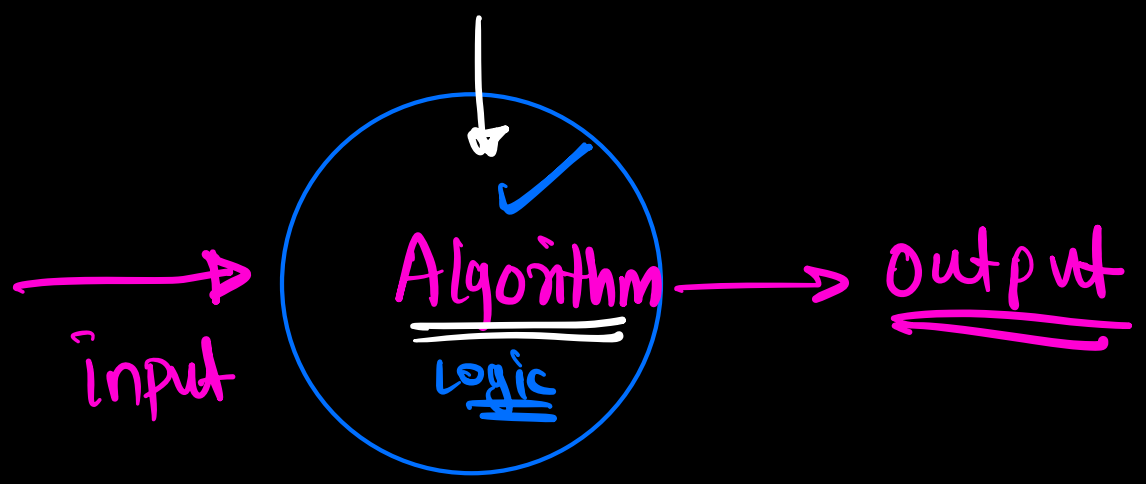
Algorithm

↳ may take 0 / more inputs

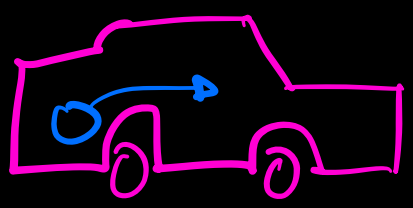
But every algorithm must produce atleast 1 output.

```
Algo( )
{
  return 10
}
```

```
Algo Square(n)
{
  return n*n
}
```



Abstraction



- ✓ A → Accelerate
- ✓ B → Break
- ✗ C → Clutch.

THANK - YOU